

WEEK 1 - VUE.JS

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COURSE OUTLINE

- ◉ Why do we need framework ?
- ◉ What is a (JS) framework?
- ◉ Library vs. framework
- ◉ Advantages & Disadvantages

WHY DO WE NEED FRAMEWORK ?

- ◉ Think of building web apps like building a house,
 - you could create all the materials from scratch and start building without any schematics, but that approach would be time-consuming and the house is probably not very good (e.g., not safe)
 - A better approach will be purchasing building materials such as wood and bricks, and then assemble them based on a well-designed blueprint

- ◉ Coding is very similar.
- ◉ When you set out to code a web app, you could code every aspect from scratch, but there are certain common features that make more sense to apply from a template if you need a wheel, for instance, it's a lot more sensible to buy one than it is to reinvent it. And that's where JavaScript frameworks come into play

WHAT IS A (JS) FRAMEWORK?

- ◉ JS frameworks are collections of JavaScript code libraries that provide developers with pre-written JS code to use for routine programming features and tasks a framework to build web app around.

LIBRARY VS. FRAMEWORK

- If you need a photo carousel in a web app, you can use a JavaScript library that already implements this .
 - This does not affect how you build the web app Just like the bricks when you build a house A framework provides a structure that organizes the parts of your app Like the blue print when building a house

ADVANTAGES & DISADVANTAGES

- ◉ Advantages: your code will be neatly structured, and the framework will provide ready-made solutions for common coding problems.
 - The people creating the framework are usually very experienced (such as the developers from Google and Facebook)
 - Like the architects that create house designs.
- ◉ Disadvantages: the imposed structure can be limiting, depends on the framework.
 - While some frameworks are designed to be all or nothing, other frameworks, such as the Vue.js we cover, allow for incremental use. This means you can apply the framework to as much or as little of your web app as you want.

WHAT ARE THE POPULAR JS FRAMEWORKS?



- ◉ Probably the most discussed topic in web development;
 - They are all very good; Well documented, mature, large user base, active development, etc.
 - They are getting similar: Share similar design concepts
 - Learning one makes it easier to start the next

- Angular (Google)
 - Started as AngularJS,
 - which is very restrictive;
 - a deeper learning curve (requires TypeScript)
 - becomes just Angular and more React like
- React (Facebook)
 - Incremental and modular
 - Medium learning curve
 - Most number of jobs now
- Vue (Evan You)
 - Combine the advantages of Angular and React
 - Easy to learn
 - Fastest growing

LEARNING RESOURCES

- ◉ Reading list:

<https://rl.talis.com/3/mdx/lists/FC2DF048-D1E6-C94D-D8FC-5261CB018FE5.html>

- Getting to know Vue.js: learn to build single page applications in Vue from scratch More details
- The official Vue.js guide: More hands on Lecture will mostly follow this one

WHAT IS VUE.JS?

- ◉ Vue progressive framework for building user interfaces.
- ◉ Vue is designed from the ground up to be incrementally adoptable.
 - The core library is focused on the view layer only, and is easy to pick up and integrate with other libraries or existing projects.
- ◉ Vue is also perfectly capable of powering sophisticated Single-Page Applications when used in combination with and modern tooling (<https://vuejs.org/v2/guide/single-le-components.html>) supporting libraries (<https://github.com/vuejs/awesome-vue#components-libraries>)

LET'S GET STARTED

Listing 1-1. Empty HTML File

```
<html lang="en">
<head>
<title>Getting to Know Vue.js</title>
</head>
<body>

</body>
</html>
```

- ◉ Add a script tag referencing the Content Delivery Network (CDN) for the library to your page and you are ready to get going!

To take this empty HTML file to a working Vue.js app, we need to add three things:

- An HTML element where we “mount” our app
- A `<script>` reference to Vue.js on the CDN
- A `<script>` element in which we create our app

Listing 1-2. The Structure of Our HTML Page

```
<!-- Div to Mount App -->
<div id="app">

</div>

<!-- Reference to Vue.js library -->
<script src="https://cdn.jsdelivr.net/npm/vue/dist/vue.js"></script>

<!-- Script Element for our first App -->
<script>

</script>
```

- ◉ That's all the setup we need before we create our first app. The next step is to add some template syntax to our app's to bind some data to it.
- ◉ For this first app, we will use what is commonly called mustache syntax. It consists of two curly braces surrounding the property name we want to inject the data from, such as in `{{ propertyName }}`. This will make our app's look like Listing 1-3.

Listing 1-3. The HTML Template for Our App

```
<!-- Div to Mount App -->  
<div id="app">  
    {{ propertyName }}  
</div>
```

In the empty `<script>` element we created, we are going to add a new instance of Vue.js, called `new Vue()`. Calling `new` on `Vue` without passing in an options object will not get us off to a good start. Therefore, we should at least tell it where to mount the app and give it a little data.

To tell our instance of Vue.js where to mount the options object, we pass in a property called `el`. The value for this will be the CSS selector. In our case, that is `#app` since we gave our `<div>` an id of `app`.

Listing 1-4. Our First Vue App!

```
var app = new Vue({
  el: '#app',
  data: {
    propertyName: 'Hello from Getting to Know Vue.js!'
  }
});
```

LET'S DO TOGETHER

Getting Started

To get started, [create an index.html file](#)

<https://gist.githubusercontent.com/chrisvfritz/7f8d7d63000b48493c336e48b3db3e52/rav>
and include Vue with:

```
In [ ]: <!-- development version, includes helpful console warnings -->  
        <script src="https://cdn.jsdelivr.net/npm/vue/dist/vue.js"></script>
```

or:

```
In [ ]: <!-- production version, optimized for size and speed -->  
        <script src="https://cdn.jsdelivr.net/npm/vue"></script>
```

- You need to be online to load the vue file.
- It is possible to install Vue locally on your computer, so you don't need to be online to use it
 - We will cover this later, together with installing other Vue tools.

index.html

```
Index.html x
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>My first Vue app</title>
5
6    <script src="https://cdn.jsdelivr.net/npm/vue/dist/vue.js"></script>
7  </head>
8  <body>
9    <div id="app">
10     <p>    {{ message }}</p>
11
12    </div>
13
14    <script>
15      var app = new Vue({
16        el: '#app',
17        data: {
18          message: 'Welcome to the Module CST3144',
19
20        }
21      })
22
23      //app.message ="I have changed the data"
24    </script>
25  </body>
26  </html>
```

Declarative Rendering

At the core of Vue.js is a system that enables us to declaratively render data to the DOM using straightforward template syntax:

```
In [ ]: <div id="app">
        {{ message }}
      </div>
```

```
In [ ]: var app = new Vue({
        el: '#app',
        data: {
          message: 'Hello Vue!'
        }
      })
```

- This is our very first Vue app!
- This looks pretty similar to rendering a string template,
- but Vue has done a lot of work under the hood.

- The data and the DOM are now linked, and everything is now **reactive**.
 - Any change to the data will be reflected in the DOM instantly, without having to refresh the page
 - The 'react' in React framework means exactly the same thing.
- Open your browser's JavaScript console
 - and set `app.message` to a different value.
 - You should see the rendered example above update accordingly.

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>My first Vue app</title>
5
6   <script src="https://cdn.jsdelivr.net/npm/vue/dist/vue.js"></script>
7 </head>
8 <body>
9   <div id="app">
10     {{ message }}
11   </div>
12
13   <script>
14     var app = new Vue({
15       el: '#app',
16       data: {
17         message: 'Hello Vue!'
18       }
19     })
20     app.message = "I have changed the data"
21   </script>
22 </body>
23 </html>
```

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Hello Vue